

Military helicopters

Combat between jet fighters was not the only innovation of the Korean War. Another of its novelties was the first extensive military use of helicopters. They were not used offensively, but still had an immediate impact, primarily as a means of evacuating wounded troops from the battlefield. Sikorsky H-5s, Bell H-13s, and Hiller H-23s carried the wounded in panniers attached to the helicopter fuselage. Receiving speedy medical attention at Mobile Army Surgical Hospitals (MASH) radically improved a wounded

soldier's chances of survival. Many men had helicopters to thank for saving their lives. Many downed airmen also owed the helicopter a debt of gratitude, for it was US Navy HO3Ss and USAF H-5s and H-19s that carried out combat rescue missions when aircraft were shot down behind communist lines or ditched in the ocean.

These errands of mercy did not exhaust the helicopters' usefulness. H-19 transport helicopters ferried troops and cargo, and helicopters were also employed as airborne command posts and for aerial observation of the battlefield.

At the same time that the Americans were discovering some of the uses of the helicopter in conventional warfare, the British were flying a version of the Sikorsky H-5, which they called the Dragonfly, in counterinsurgency operations against communist guerrillas in Malaya – a foretaste of the prominent role that helicopters would later play in Vietnam.

The Korean War ended in a ceasefire in 1953, leaving Korea divided as it had been before the fighting started. The evidence from this "limited war" suggested that, at least when prevented by

EVOLUTION OF THE HELICOPTER

AIRCRAFT DESIGNER IGOR SIKORSKY said that "the idea of a vehicle that could lift itself vertically from the ground and hover motionless in the air was probably born at the same time that man first dreamed of flying." It was certainly an option explored, by Sikorsky among others, as early as the opening decade of the 20th century. Frenchman Paul Cornu is often credited with getting the first helicopter briefly off the ground in 1907, but it would be a long time before the goal of controlled, manned, sustained rotor flight was attained.

Of the many problems confronting helicopter designers, the most intractable was control. For a start, the torque generated by the rotating blade – the helicopter's equivalent of the airplane's wing – would automatically make the helicopter spin in the opposite direction to the blade. In 1912, Russian experimenter Boris Yuriev showed that the torque could be overcome by mounting a smaller vertical propeller on the tail, but his work was largely ignored.

Even if the helicopter did not spin like a top, it was hard to figure out how it was to be made to rise and fall, move backward or forward, change direction, or hover. Groundbreaking progress on this point was made by Argentinian engineer Raul Pateras de Pescara, who figured out how to vary the pitch of each rotor blade so that the helicopter would tilt in different directions. This "cyclic control" would allow a pilot to fly the aircraft forward or backward, left or right.

Further progress with rotors came from Spanish engineer Juan de la Cierva's 1920s invention, the Autogiro. Like an airplane, the Autogiro was driven by a propeller, but lift was provided by rotors, which were spun around by the air as the aircraft moved forward. To make this hybrid vehicle work, De la Cierva had to exhaustively explore the properties of spinning rotors. The example of his machines gave a new impetus to helicopter research.

By the second half of the 1930s, French, German and American experimenters were all vying to produce a practical helicopter. Frenchman Louis Breguet is generally accepted to have won the race in 1935, but Heinrich Focke significantly improved with his design with the Fa 61 the following year. In the United States, in 1939 Sikorsky returned to the experiments he had abandoned almost 30 years earlier, producing the VS-300. Factory production of helicopters started during World War II.

Sikorsky emerged as the individual publicly identified with the success of this new form of aviation. His ultimate dream was that the helicopter



CORNU'S HELICOPTER

Frenchman Paul Cornu claimed to have risen from the ground for a few seconds in his primitive twin-rotor helicopter in November 1907. Modern studies suggest that it could not have sustained its own weight and that of a pilot.

would become the successor to the automobile, parked on every front lawn in America. It did not turn out like that, but given the many and varied uses rotating-wing craft have found, this hardly counts as a disappointment.

CIERVA C-30 AUTOGIRO

Juan de la Cierva's Autogiros were the first practical rotary-wing aircraft. Since the rotor was unpowered it provided lift but not propulsion. The Autogiro shown here is a C-30 built under license by Avro for the RAF as a Rota 1.

